

Lab 12 - Malware Sandbox Analysis

SOC Analyst Portfolio Lab

Objective

Analyze a public malware sandbox report without executing real malware locally. The investigation focuses on process behavior, credential-stealing activity, registry persistence, network communication, IOC extraction, MITRE ATT&CK mapping, and SOC triage.

Environment

Component	Use in Lab
Public malware sandbox report	Safe source of behavioral telemetry for the Agent Tesla sample
Analyst workstation	Used only to review the public report; no malware was downloaded or executed
Agent Tesla sample	Information-stealing malware analyzed through sandbox evidence
MITRE ATT&CK	Framework used to map observed behaviors to defensive investigation concepts

Scope and Safety

No malware sample was downloaded or executed on the analyst workstation or virtual machines. All findings were derived from the public sandbox report and screenshots captured during the analysis.

Sample and Process Evidence

Evidence	Observed Result	Analyst Meaning
Main sample	AgentTesla.exe	Primary malware sample shown in the sandbox report

Sample path	C: \Users\admin\AppData\Local\Temp\AgentTesla.exe	Execution from a user temporary directory is notable in malware analysis
Suspicious child process	RegSvcs.exe	Legitimate Microsoft .NET utility launched as a child of AgentTesla.exe
RegSvcs path	C: \Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe	Confirms the child process used a legitimate Windows/.NET binary

Process description

Process information

PID	CMD	Path	Indicators	Parent process
2448	"C:\Users\admin\AppData\Local\Temp\AgentTesla.exe"	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe		AgentTesla.exe

Information

User: admin Company: Microsoft Corporation
Integrity Level: MEDIUM Description: Microsoft .NET Services Installation Utility
Version: 4.8.3761.0 built by: NET48REL1

Modules

Images

- c:\windows\microsoft.net\framework\v4.0.30319\regsvcs.exe
- c:\windows\system32\ntdll.dll
- c:\windows\system32\kernel32.dll
- c:\windows\system32\kernelbase.dll
- c:\windows\system32\ole32.dll
- c:\windows\system32\msvcrt.dll
- c:\windows\system32\gdi32.dll
- c:\windows\system32\user32.dll
- c:\windows\system32\lpk.dll
- c:\windows\system32\usp10.dll

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Figure 1. Sandbox process information showing AgentTesla.exe launching RegSvcs.exe.

Behavioral Findings

The sandbox identified a combination of malicious and suspicious behaviors consistent with an information-stealing trojan. The most important findings are summarized below.

Behavior	Assessment
Steals credentials from web browsers	Strong evidence of credential-access behavior
Reads browser cookies	Supports collection of browser session or account-related data
Accesses Microsoft Outlook profiles	Potential collection of email account/profile information
Checks external IP	Host/network discovery behavior
Reads computer name, machine GUID, language, and environment values	System discovery and host fingerprinting
Changes autorun registry value	Persistence behavior using a Windows logon autorun location
Connects to FTP and unusual ports	Suspicious outbound communication consistent with possible exfiltration or command-and-control

Behavior activities

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MALICIOUS

STEALER has been found (auto)

- AgentTesla.exe (PID: 2452)

Steals credentials from Web Browsers

- RegSvc.exe (PID: 2448)

AGENTTESLA has been detected (SURICATA)

- RegSvc.exe (PID: 2448)

Stealers network behavior

- RegSvc.exe (PID: 2448)

AGENTTESLA has been detected (YARA)

- RegSvc.exe (PID: 2448)

Actions looks like stealing of personal data

- RegSvc.exe (PID: 2448)

Changes the autorun value in the registry

- RegSvc.exe (PID: 2448)

SUSPICIOUS

Reads settings of System Certificates

- RegSvc.exe (PID: 2448)

Executable content was dropped or overwritten

- RegSvc.exe (PID: 2448)

Reads the Internet Settings

- RegSvc.exe (PID: 2448)

Checks for external IP

- RegSvc.exe (PID: 2448)

Connects to FTP

- RegSvc.exe (PID: 2448)

Reads browser cookies

- RegSvc.exe (PID: 2448)

Connects to unusual port

- RegSvc.exe (PID: 2448)

The process connected to a server suspected of theft

- RegSvc.exe (PID: 2448)

Accesses Microsoft Outlook profiles

- RegSvc.exe (PID: 2448)

Process drops legitimate windows executable

- RegSvc.exe (PID: 2448)

INFO

Reads mouse settings

- AgentTesla.exe (PID: 2452)

The sample compiled with english language support

- AgentTesla.exe (PID: 2452)

- RegSvc.exe (PID: 2448)

Checks supported languages

- RegSvc.exe (PID: 2448)

- AgentTesla.exe (PID: 2452)

Reads the computer name

- RegSvc.exe (PID: 2448)

Reads the machine GUID from the registry

- RegSvc.exe (PID: 2448)

Reads the software policy settings

- RegSvc.exe (PID: 2448)

Disables trace logs

- RegSvc.exe (PID: 2448)

Creates files or folders in the user directory

- RegSvc.exe (PID: 2448)

Reads Environment values

- RegSvc.exe (PID: 2448)

ULTRAVNC has been detected

- RegSvc.exe (PID: 2448)

.NET Reactor protector has been detected

- RegSvc.exe (PID: 2448)

Create files in a temporary directory

- AgentTesla.exe (PID: 2452)

Figure 2. Sandbox behavior summary showing credential theft, persistence, discovery, and suspicious network activity.

Registry Persistence

The report provided direct evidence of logon persistence through a Windows Run key. This is stronger persistence evidence than generic registry writes because the Run location can launch the referenced executable when the user signs in.

Field	Observed Value
Registry key	HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
Value name	sgxlb
Executable	C:\Users\admin\AppData\Roaming\sgxlb\sgxlb.exe
Assessment	Configured for automatic execution when the user logs on

Registry activity				☒ Add for printing
Total events	Read events	Write events	Delete events	
3 623	3 596	27	0	
Modification events				
(PID) Process: (2448) RegSvcs.exe Operation: write Value: 1048576	Key: HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Tracing\RegSvcs_RASMANCS Name: MaxFileSize			
(PID) Process: (2448) RegSvcs.exe Operation: write Value: %windir%\tracing	Key: HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Tracing\RegSvcs_RASMANCS Name: FileDirectory			
(PID) Process: (2448) RegSvcs.exe Operation: write Value: en-US	Key: HKEY_CLASSES_ROOT\Local Settings\MuiCache\182\52C64B7E Name: LanguageList			
(PID) Process: (2448) RegSvcs.exe Operation: write Value: C:\Users\admin\AppData\Roaming\sgxlb\sgxlb.exe	Key: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run Name: sgxlb			
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Figure 3. Registry activity showing the Run-key persistence entry for sgxlb.exe.

Network Activity

The report showed 11 TCP/UDP connections, 3 DNS requests, 12 threat detections, and no HTTP requests. The most relevant outbound connections were generated by RegSvcs.exe.

Destination	Port	Assessment
api.ipify.org / 104.26.13.205	443	Legitimate public-IP service; suspicious in this context because malware can use it to learn the victim public IP
ftp.haliza.com.my / 110.4.45.197	21	Suspicious FTP control-channel connection from an information-stealing sample
ftp.haliza.com.my / 110.4.45.197	High ports	Additional FTP-related connections consistent with data-channel style behavior

Network activity

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HTTP(S) requests

0

TCP/UDP connections

11

DNS requests

3

Threats

12

HTTP requests

No HTTP requests

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Connections

PID	Process	IP	Domain	ASN	CN	Reputation
—	—	224.0.0.252:5355	—	—	—	whitelisted
4	System	192.168.100.255:137	—	—	—	whitelisted
4	System	192.168.100.255:138	—	—	—	whitelisted
1080	svchost.exe	224.0.0.252:5355	—	—	—	whitelisted
2448	RegSvcs.exe	104.26.13.205:443	api.ipify.org	CLOUDFLARENET	US	shared
2448	RegSvcs.exe	110.4.45.197:21	ftp.haliza.com.my	Exa Bytes Network Sdn.Bhd.	MY	whitelisted
2448	RegSvcs.exe	110.4.45.197:55568	ftp.haliza.com.my	Exa Bytes Network Sdn.Bhd.	MY	whitelisted
2448	RegSvcs.exe	110.4.45.197:62032	ftp.haliza.com.my	Exa Bytes Network Sdn.Bhd.	MY	whitelisted
2448	RegSvcs.exe	110.4.45.197:60821	ftp.haliza.com.my	Exa Bytes Network Sdn.Bhd.	MY	whitelisted
2448	RegSvcs.exe	110.4.45.197:65491	ftp.haliza.com.my	Exa Bytes Network Sdn.Bhd.	MY	whitelisted

Figure 4. Network activity showing RegSvcs.exe connections to api.ipify.org and ftp.haliza.com.my.

IOC and Reputation Analysis

Reputation labels were treated as context rather than final verdicts. A 'shared' service or 'whitelisted' domain can still be relevant when malware uses it during execution.

IOC Type	Value	Analyst Note
Filename	AgentTesla.exe	Primary malware sample name
Persistence executable	sgxlb.exe	Referenced by the Run-key persistence entry
Domain	api.ipify.org	Legitimate/shared service; suspicious only in the malware execution context
Domain	ftp.haliza.com.my	Whitelisted reputation label, but suspicious in this execution context because the malware used FTP
IP	104.26.13.205	Resolved address for api.ipify.org; shared infrastructure and not inherently malicious
IP	110.4.45.197	Resolved address for ftp.haliza.com.my; relevant to observed FTP activity
Registry key	HKCU\Software\Microsoft\Windows\CurrentVersion\Run	Persistence indicator

MITRE ATT&CK Mapping

Technique / Tactic	Application in This Lab
T1547.001 - Registry Run Keys / Startup Folder	Run-key modification pointing to sgxlb.exe supports logon persistence
Credential Access	Sandbox reported theft of browser credentials
Collection	Browser cookies and Outlook profile access support collection behavior
Discovery	Public-IP lookup, computer name, machine GUID, language,

	and environment checks
Exfiltration / C2-related behavior	FTP communication is consistent with suspicious outbound transfer behavior, but the report does not prove the exact data transferred

SOC Analyst Triage

Field	Assessment
Severity	High
Verdict	Malicious - credential-stealing malware behavior
Escalation	Yes
Reasoning	Credential theft, browser/Outlook data access, persistence through a Run key, system discovery, and suspicious FTP communication together support a high-confidence malicious assessment.

Recommended SOC Response

- Isolate the affected endpoint from the network.
- Contain malicious processes and preserve forensic evidence.
- Block confirmed malicious domains and destinations where appropriate.
- Reset potentially exposed credentials and review MFA status.
- Review browser, email, and FTP account activity for unauthorized access.
- Hunt across the environment for the same file names, persistence key, domains, and related processes.
- Review sign-in logs and endpoint telemetry for evidence of follow-on compromise.

Investigation Lessons

- Behavioral evidence should carry more weight than a single reputation label.
- Legitimate services such as api.ipify.org can still be used by malware for discovery.

- A whitelisted domain can still be suspicious when the process context and protocol usage are abnormal.
- Direct Run-key modification provides strong persistence evidence and maps cleanly to T1547.001.
- FTP activity from an information stealer is suspicious, but analysts should avoid claiming confirmed exfiltration unless the transferred data is evidenced.
- Public sandbox analysis is a safe way to investigate real malware behavior without executing malware locally.

Analyst Conclusion

The public sandbox report provided strong evidence that the analyzed Agent Tesla sample behaved as an information stealer. The most significant findings were browser credential theft, browser-cookie and Outlook-profile access, Run-key persistence, host discovery, and suspicious FTP communication. Reputation labels such as 'whitelisted' or 'shared' were treated as context rather than final verdicts; behavioral evidence remained the primary basis for the assessment.

Safety note: This lab was completed by reviewing a public sandbox report only. No malware was executed locally.